

Project Close-out Report

Anti-Front-Running ZKPerps on Cardano w/ Midnight ZK

Project Name	Anti-Front-Running ZKPerps on Cardano w/ Midnight ZK
Project Number	#1400010
Fund / Challenge	Project Catalyst Fund 14 — Cardano Use Cases: Concept
Project Manager	Sricharan Ganta
Project URL	projectcatalyst.io — Fund 14 Proposal
Date Started	November 24, 2025
Date Completed	May 13, 2026

Challenge KPIs and How the Project Addressed Them

The project focused on building privacy-preserving decentralized perpetual derivatives infrastructure on Cardano using Midnight Zero-Knowledge proofs.

Open-Source Infrastructure Delivery

The project successfully delivered a fully open-source Anti-Front-Running ZK Perpetuals protocol integrating Zero-Knowledge proof generation and on-chain verification across five Midnight Compact contracts, Cardano L1 settlement anchoring via Aiken PlutusV3 validators, commitment-reveal order placement with SHA-256 and 128-bit secret nonces, and full marketplace backend and frontend infrastructure. All code, smart contracts, APIs, documentation, and integration workflows were publicly released under the MIT License.

Privacy-Preserving DeFi Infrastructure

The project demonstrated how Zero-Knowledge proofs and cryptographic commitment schemes can prevent front-running, sandwich attacks, and order information leakage in decentralized perpetual derivatives. The architecture was successfully validated across Cardano Preprod and Midnight Preview environments with 9 recorded testnet transactions.

Cross-Chain and Layered Architecture Validation

The project demonstrated interoperability between Cardano Layer 1, Midnight privacy-preserving smart contracts, and off-chain commitment and encryption systems. Successful execution of ZK proof workflows, settlement anchoring, matching, liquidation, and aggregated proof bundling proved the feasibility of advanced privacy-preserving DeFi applications on Cardano infrastructure.

Project KPIs and How the Project Addressed Them

Software Requirements and Technical Design

A complete Software Requirements Specification (SRS) and Architecture Document were produced covering ZK commitment-reveal order placement, five-contract Midnight Compact pipeline, Cardano L1 settlement anchoring, front-running prevention threat model, privacy enforcement validation criteria, and benchmark targets. The project successfully validated the implementation of all five Midnight contracts, Cardano settlement anchors, front-running prevention across four attack vectors, and privacy enforcement across eight validation checks.

Prototype Development and Testnet Validation

The project deployed and tested core infrastructure across Cardano Preprod and Midnight Preview environments. Validated workflows included private order commitment creation, five-contract ZK proof pipeline deployment (blocks 46574–46611), `proveTraderOrderAuthority`, `bindL1SettlementAnchor`, `proveAndFinalizeMatch`, `proveSettlementTransition`, `proveLiquidationBreach`, and `proveAggregatedProofBundle`. The system demonstrated end-to-end trader and settlement workflows through integrated testnet environments.

Functional MVP Deployment and Open-Source Release

The final milestone delivered a fully functional MVP deployment of the Anti-Front-Running ZKPerps protocol on Cardano testnet. The platform demonstrated private order commitment creation, five-contract Midnight ZK proof pipeline execution, Cardano L1 settlement anchoring via PlutusV3, front-running prevention validation (4 attacks blocked), and privacy enforcement validation (8 checks passed).

The MVP included a live frontend marketplace UI, backend APIs with Swagger/OpenAPI documentation, Cardano + Midnight integrated workflows, and a public MIT-licensed open-source release.

Live MVP (UI + backend, same origin)	flour-provisions-ipod-indicating.trycloudflare.com
Main Repository	github.com/Nucastio/Anti-Front-Running-ZKPerps-on-Cardano-w-MidnightZK

Useful paths on the live origin:

/	Marketplace UI
/health	Service health probe
/docs	Swagger UI (interactive API explorer)
/api/openapi.json	Machine-readable OpenAPI spec
/api/evidence	Live Blockfrost evidence stream
/api/cardano/verify-anchor?txHash=...	Verify any Cardano settlement-anchor tx
POST /api/pipeline/run-order	Kick off the five-contract pipeline (returns runId)
GET /api/pipeline/runs/{runId}	Poll pipeline execution status

Key Achievements

- Delivered one of the first open-source implementations combining Cardano smart contracts, Midnight privacy infrastructure, Zero-Knowledge verification, and cryptographic front-running prevention for decentralized perpetual derivatives.
- Successfully deployed and verified five Midnight Compact contracts (order, matching, settlement, liquidation, aggregate) across 37 blocks with 12 recorded ZK transactions.
- Demonstrated front-running prevention across four distinct attack vectors (order sniping, price front-running, MEV reordering, sandwich attacks) — all cryptographically blocked.
- Validated privacy enforcement across eight checks confirming price, size, trader identity, and order metadata are never exposed on-chain.
- Anchored settlement finality on Cardano L1 via Aiken PlutusV3 settlement_anchor validator with inline AnchorDatum on Preprod testnet.
- Released complete backend systems, frontend marketplace UI, smart contracts, ZK infrastructure, and developer documentation under the MIT License.

Impact

Technical and ecosystem impact:

- Successful completion of all funded milestones
- Functional MVP deployment on Cardano testnet
- 9 recorded testnet transactions (7 Midnight + 2 Cardano Preprod)
- Five Midnight Compact contracts deployed and verified
- Delivery of 12 public technical documentation resources
- Public deployment of live backend and frontend MVP

Challenges addressed in DeFi and decentralized systems:

- Front-running and MEV prevention in decentralized derivatives
- Privacy-preserving order placement and execution
- Secure settlement finality with cross-chain anchoring
- Zero-Knowledge proof integration for DeFi applications
- Decentralized perpetual derivatives infrastructure

Why is this Project Important?

This project demonstrates how Cardano and Midnight can support a new class of privacy-preserving decentralized finance applications centered around fair order execution, front-running prevention, and secure settlement infrastructure.

As DeFi ecosystems continue to grow, there is increasing demand for front-running and MEV-resistant order execution, privacy-preserving derivatives trading, secure cross-chain settlement mechanisms, Zero-Knowledge proof integration for financial applications, and tamper-resistant proof systems for order integrity. The Anti-Front-Running ZKPerps protocol addresses these challenges through the integration of Zero-Knowledge verification across five Midnight Compact contracts, commitment-reveal order schemes, AES-256-GCM encrypted shielded order pools, Cardano L1 settlement anchoring, and aggregated proof bundling.

The project proves that advanced privacy-preserving DeFi applications can be implemented on Cardano while maintaining security, interoperability, auditability, and decentralized settlement principles. The outcomes establish a strong foundation for future decentralized derivatives platforms, privacy-focused DeFi infrastructure, and next-generation Web3 trading systems.

Links to Other Relevant Project Sources or Documents

Project Repository	github.com/Nucastio/Anti-Front-Running-ZKPerps-on-Cardano-w-MidnightZK
Project Catalyst Milestones	milestones.projectcatalyst.io/projects/1400010
Software Requirements Spec	docs/SRS.md
Architecture Document	docs/architecture.md
Testnet Evidence	docs/testnet-evidence.md
Benchmarks	docs/benchmarks.md
Documentation Repository	docs/ (12 documents)
Live MVP (UI + backend)	flour-provisions-ipod-indicating.trycloudflare.com
Live MVP API (Swagger)	flour-provisions-ipod-indicating.trycloudflare.com/docs
Live MVP OpenAPI spec	flour-provisions-ipod-indicating.trycloudflare.com/api/openapi.json
Live evidence stream	flour-provisions-ipod-indicating.trycloudflare.com/api/evidence
Demo Video (UI MVP Walkthrough)	demo-zkperps-ui.mp4

Project Close-out Video

https://youtu.be/MxLBY_7tuw0